C. Those Who Are With Us

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1200 | Tags: greedy, implementation | https://codeforces.com/problemset/problem/2121/C

You are given a matrix of integers with n rows and m columns. The cell at the intersection of the i-th row and the j-th column contains the number a_{ij}.

You can perform the following operation **exactly once**:

- Choose two numbers $1 \le r \le n$ and $1 \le c \le m$.
- For all cells (i, j) in the matrix such that $i = r$ or $j = c$, decrease a_{ij} by one.

You need to find the minimal possible maximum value in the matrix a after performing exactly one such operation.

Input

Each test consists of multiple test cases. The first line contains a single integer t ($1 \le t \le 10^4$) — the number of test cases. The description of the test cases follows.

The first line of each test case contains two integers n and m ($1 \le n \cdot m \le 10^5$) — the number of rows and columns in the matrix.

The next n lines of each test case describe the matrix a. The i-th line contains m integers $a_{i1}, a_{i2}, \ldots, a_{im}$ ($1 \le a_{ij} \le 100$) — the elements in the i-th row of the matrix.

It is guaranteed that the sum of $n \cdot m$ across all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, output the minimum maximum value in the matrix a after performing exactly one operation.

Example

Input

```
10
1 1
1
1 2
1 2
2 1
2
1
2 2
4 2
3 4
3 4
1 2 3 2
3 2 1 3
2 1 3 2
4 3
1 5 1
3 1 3
5 5 5
3 5 1
4 4
1 3 3 2
2 3 2 2
1 2 2 1
3 3 2 3
2 2
2 2
1 2
3 2
1 2
2 1
1 2
3 3
2 1 1
1 2 1
1 1 2
```

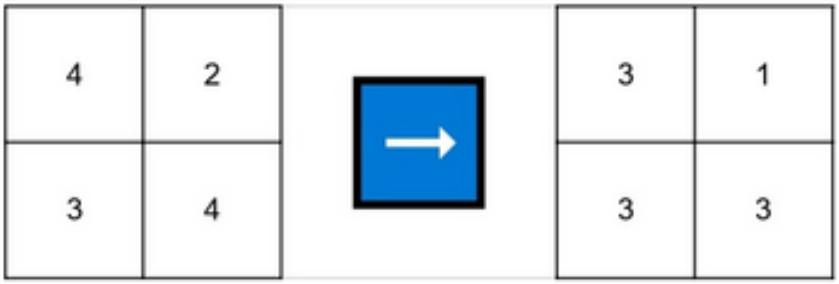
Output

```
0
1
1
3
2
4
3
1
1
2
```

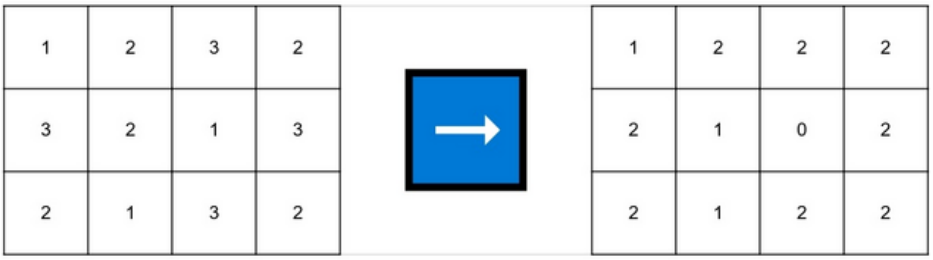
Note

In the first three test cases, you can choose $r = 1$ and $c = 1$.

In the fourth test case, you can choose $r = 1$ and $c = 2$.



In the fifth test case, you can choose $r = 2$ and $c = 3$.



In the sixth test case, you can choose $r = 3$ and $c = 2$.